

HYPOTHESIS TESTING

P-Values

T-Test

Z-Test

Confidence Intervals

Error Types



What is Hypothesis Testing?

 *Think of it like a court trial: You assume someone is innocent (null hypothesis) until proven guilty (alternative hypothesis). The data is your evidence.*

1

Make a claim

Start with an assumption you want to test (e.g., 'This medicine works')

2

Collect evidence

Gather data through experiments or surveys

3

Decide

Use statistics to accept or reject your initial assumption

Null & Alternative Hypotheses

H_0 Null Hypothesis

The default assumption — nothing new is happening.

- Assumes no effect, no difference
- Always includes an equals sign ($=$, $\leq$, $\geq$)
- We try to disprove this

Example: H_0 : The new drug has no effect on blood pressure.

H_1 Alternative Hypothesis

What you believe is actually true — the new claim.

- Proposes an effect or difference exists
- Uses $\neq$, $>$, or $<$ signs
- This is what we want to prove

Example: H_1 : The new drug reduces blood pressure.

The P-Value Explained

The p-value is the probability of getting your results (or something more extreme) IF the null hypothesis were true.

The Decision Rule

$p \leq 0.05 \rightarrow \text{Reject } H_0$ ✓

Strong evidence against null hypothesis

$p > 0.05 \rightarrow \text{Fail to Reject } H_0$ ✗

Not enough evidence to reject null

Real Life Example

You flip a coin 100 times and get 65 heads.

H_0 : The coin is fair (50% heads)

If p-value = 0.002 $\rightarrow$ This result is very unlikely with a fair coin
 $\rightarrow$ Reject H_0 !

Conclusion: The coin is likely rigged.

Type I & Type II Errors

	H_0 is TRUE	H_0 is FALSE
Reject H_0	Type I Error (α) False Positive 'Convicting an innocent person'	✓ Correct Decision Power = $(1 - \beta)$
Don't Reject H_0	✓ Correct Decision Confidence = $(1 - \alpha)$	Type II Error (β) False Negative 'Freeing a guilty person'

💡 α = significance level (usually 0.05) | β = probability of Type II error | Power = $1 - \beta$ (ability to detect true effects)

Z-Test vs T-Test

Z-Test

When to use: Large sample ($n \geq 30$) OR population std dev (σ) is known

Formula: $Z = (\bar{x} - \mu) / (\sigma / \sqrt{n})$

Analogy: Like knowing the exact weight of ALL apples in a farm

Example: Testing if avg height of 200 students = 170cm (σ known)

VS

T-Test

When to use: Small sample ($n < 30$) AND σ is unknown (use sample std dev 's')

Formula: $t = (\bar{x} - \mu) / (s / \sqrt{n})$

Analogy: Estimating when you only have a handful of apples

Example: Testing if avg score of 15 students = 70 (σ unknown)

Confidence Intervals

A confidence interval gives a range of plausible values for the true population parameter. A 95% CI means: if we repeated this study 100 times, 95 of the intervals would contain the true value.

Formula

$$CI = \bar{X} \pm Z^* \times (\sigma / \sqrt{n})$$

Critical Values (Z*)

90% CI $Z^* = 1.645$

95% CI $Z^* = 1.960$

99% CI $Z^* = 2.576$



Worked Example

A survey of 100 students finds average study time $\bar{x} = 6$ hrs/day, $\sigma = 2$ hrs.

Build a 95% CI:

$$\begin{aligned} CI &= 6 \pm 1.96 \times (2 / \sqrt{100}) \\ &= 6 \pm 1.96 \times 0.2 \\ &= 6 \pm 0.392 \end{aligned}$$



CI = (5.608, 6.392) hours

Margin of Error

 Like a TV poll: 'Candidate A leads with 52% ± 3%' — that ±3% is the Margin of Error. It tells you how far off your estimate might be.

$$\text{MOE} = Z^* \times (\sigma / \sqrt{n})$$

Example: $n=400$, $\sigma=10$, 95% CI
 $\text{MOE} = 1.96 \times (10 / \sqrt{400}) = 1.96 \times 0.5 = 0.98$
So estimate is within ± 0.98 of truth



Confidence Level ↑

MOE increases



Sample Size ↑

MOE decreases



Std Deviation ↑

MOE increases

Key Takeaways

Hypotheses

H_0

H_0 = default assumption; H_1 = what you want to prove

P-Value

p

$p \leq 0.05 \rightarrow$ reject null hypothesis (result is significant)

Error Types



Type I (false positive) and Type II (false negative) errors

Tests

Z/t

Z-test for large n or known σ ; T-test for small n or unknown σ

Confidence Interval

CI

Range likely containing the true parameter value

Margin of Error

$\pm$

How precise your estimate is — larger n means smaller MOE